

Section 6.1

1. a. Let X denote the random variable of accompanying data. We use $\bar{X}$ as estimator.

$$\bar{X} = \frac{1}{n} \sum x_i = \frac{1}{27} \cdot 219.8 = 8.14$$

- b. We use the median $\tilde{X}$ as estimator.
After sorting, the data is:

5.9	6.3	6.3	6.5	6.8	6.8	7.0
7.0	7.2	7.3	7.4	7.6	7.7	7.7
7.8	7.8	7.9	8.1	8.2	8.7	9.0
9.7	9.7	10.7	11.3	11.6	11.8	

$$\tilde{X} = X_{\frac{n+1}{2}} = X_{14} = 7.7$$

- c. We use $\sqrt{s^2}$ as the estimator.

$$s^2 = \frac{1}{n-1} \left[\sum x_i^2 - \frac{(\sum x_i)^2}{n} \right]$$

$$= \frac{1}{26} \cdot \left(11860.94 - \frac{219.8^2}{27} \right)$$

$$= 2.754$$

$$\sqrt{s^2} = 1.66$$

d. We use the proportion of the sample as the estimator.

$$p = \frac{\text{the number exceeds 10 in sample}}{n}$$

$$= \frac{4}{27}$$

$$= 0.148$$

e. We use $\frac{\sqrt{s^2}}{\bar{x}}$ of sample as the estimator

$$\frac{\sqrt{s^2}}{\bar{x}} = \frac{1.66}{8.14} = 0.204$$

8. a. We use the proportion of non-defective components of sample as estimator.

$$p = \frac{80-12}{80} = 0.85$$

b. Since the estimated probability of a non-defective component is $p = 0.85$

Thus the proportion of all such system work properly is estimated as

$$P(\text{system works}) = p^2 = 0.85^2 = 0.7225$$

9. a. We use $E(\bar{x})$ as the unbiased estimator of μ .

$$E(\bar{x}) = \sum x_i p_i = 0 \times \frac{18}{150} + 1 \times \frac{37}{150} + \dots + 7 \times \frac{1}{150} \\ = 2.11$$

$$b. \sigma_{\bar{x}} = \frac{\sigma_x}{\sqrt{n}} = \sqrt{\frac{\mu}{n}} = \sqrt{\frac{E(\bar{x})}{n}} = \sqrt{\frac{2.11}{150}} = 0.12$$

$$13. E(\hat{\theta}) = 3E(\bar{x})$$

$$E(\bar{x}) = \int_{-\infty}^{\infty} x f(x) dx = 0.5 \int_{-1}^1 (x + \theta x^2) dx \\ = 0.5 \left[\frac{1}{2} x^2 + \frac{1}{3} \theta x^3 \right]_{-1}^1 \\ = \frac{1}{3} \theta$$

$$\text{Thus } E(\hat{\theta}) = 3E(\bar{x}) = \theta.$$

$\hat{\theta} = 3\bar{x}$ is an unbiased estimator of θ .

Section 6.2

20. a. The distribution is binomial distribution.

$$f(3; 20, p) = \binom{20}{3} \cdot p^3 (1-p)^{17}$$

We find the maximum of $g(p) = p^3 (1-p)^{17}$.

$$\ln g(p) = 3 \ln p + 17 \ln(1-p).$$

$$\text{Then } \frac{d}{dp} [\ln g(p)] = \frac{3}{p} - \frac{17}{1-p} = 0$$

$$\text{Thus } p = \frac{3}{20} = 0.15$$

$$\text{b. } E(\hat{p}) = E\left(\frac{X}{n}\right) = \frac{1}{n} E(X) = \frac{1}{n} \cdot np = p$$

Thus the estimator is unbiased.

c. By (a). We have $\hat{p} = 0.15$.

Thus the mle of $(1-p)^5$ is

$$(1-\hat{p})^5 = (1-0.15)^5 = 0.4437$$

21. a. The first moment is $\bar{X}$.

$$\text{Thus } \bar{X} = E(X) = \beta \cdot \Gamma(1 + 1/\alpha)$$

The second moment is $\frac{1}{n} \sum X_i^2$.

$$\begin{aligned} \text{and } E(X^2) &= [E(X)]^2 + V(X) \\ &= \beta^2 \Gamma(1 + 2/\alpha) \end{aligned}$$

$$\text{Thus } \frac{1}{n} \sum X_i^2 = \beta^2 \Gamma(1 + 2/\alpha).$$

Since $\beta = \frac{\bar{X}}{\Gamma(1 + 1/\alpha)}$, we have

$$\frac{\Gamma(1 + 2/\alpha)}{[\Gamma(1 + 1/\alpha)]^2} = \frac{\sum X_i^2}{n \bar{X}^2}$$

After solving the complicated equation involving the gamma function, we can get the estimate of α . Then by

$\beta = \frac{\bar{X}}{\Gamma(1 + 1/\alpha)}$, β can be found from a table of gamma function.

b. Since $\frac{\sum x_i^2}{n\bar{x}^2} = 1.052 \approx \frac{1}{0.95} = \frac{\Gamma(1.4)}{[\Gamma(1.2)]^2}$

Thus $1 + 2/\alpha = 1.4$. $\alpha = 5$.

Then by the table, $\Gamma(1.2) = 0.918$

Thus $\beta = \frac{\bar{x}}{\Gamma(1+1/\alpha)} = \frac{28.0}{0.918} = 30.495$

29. a. $f(x_1, x_2, \dots, x_n; \lambda, \theta)$

$$= f(x_1; \lambda, \theta) \cdot \dots \cdot f(x_n; \lambda, \theta)$$

$$= \lambda^n \cdot e^{-\lambda \sum (x_i - \theta)}$$

Let $g(x_1, \dots, x_n) = \ln f(x_1, \dots, x_n; \lambda, \theta)$

$$= n \ln \lambda - \lambda \sum (x_i - \theta)$$

Thus $\frac{\partial g(x_1, \dots, x_n)}{\partial \lambda} = \frac{n}{\lambda} - \sum (x_i - \theta) = 0$

$$\lambda = \frac{n}{\sum (x_i - \theta)} = \frac{1}{\bar{x} - \theta}. \text{ Since } x_i \geq \theta, \lambda > 0$$

and $\frac{\partial g(x_1, \dots, x_n)}{\partial \theta} = \lambda n > 0$. Thus $g(x_1, \dots, x_n)$

increases about θ .

Since $x_i \geq \theta$. $\hat{\theta} = \min\{x_1, \dots, x_n\}$ as the minimum.

Thus $\hat{\theta} = \min\{x_1, \dots, x_n\}$. $\hat{\lambda} = \frac{1}{\bar{x} - \min\{x_1, \dots, x_n\}}$

b. $\hat{\theta} = 0.64$ by the values.

$$\mu = \frac{1}{10} \sum x_i = 5.58.$$

$$\text{Thus } \hat{\lambda} = \frac{1}{\mu - \hat{\theta}} = \frac{1}{5.58 - 0.64} = 0.2024.$$

32. a. Since $Y \leq y \Leftrightarrow X_i \leq y$ for each i .

$$\begin{aligned} F_Y(y) &= P(Y \leq y) = P(x_1, x_2, \dots, x_n \leq y) \\ &= P(X_1 \leq y) \cdot P(X_2 \leq y) \cdot \dots \cdot P(X_n \leq y) \end{aligned}$$

Since X_i is uniform distribution on $[0, \theta]$

$$F(x) = \frac{x}{\theta}, \quad 0 \leq x \leq \theta.$$

$$\text{Thus } F_Y(y) = \left(\frac{y}{\theta}\right)^n, \quad (0 \leq y \leq \theta).$$

when $y < 0$, $F_Y(y) = 0$; when $y > \theta$, $F_Y(y) = 1$.

Take derivative, $f_Y(y) = \frac{ny^{n-1}}{\theta^n} \cdot 0 \leq y \leq \theta$.
 $\begin{cases} 0, \text{ otherwise.} \end{cases}$

$$\begin{aligned} \text{b. } E(\hat{\theta}) &= E(Y) = \int_{-\infty}^{\infty} y \cdot f_Y(y) \, dy \\ &= \int_0^{\theta} \frac{ny^n}{\theta^n} \, dy \\ &= \frac{1}{\theta^n} \cdot \frac{n}{n+1} [y^{n+1}]_0^{\theta} \\ &= \frac{n\theta}{n+1} \neq \theta \end{aligned}$$

Thus the mle is biased.

$$\text{Let } \hat{\theta} = \frac{n+1}{n} \cdot \max(X_i) = \frac{n+1}{n} Y.$$

$$E(\hat{\theta}) = \frac{n+1}{n} \cdot E(Y) = \frac{n+1}{n} \cdot \frac{n}{n+1} \theta = \theta.$$

Thus $\frac{(n+1)\max(X_i)}{n}$ is unbiased.

